Oxygen Sabotage Assignment

Someone set off the Oxygen.exe process and almost killed us all!!!

The adversary smashed the screen of the computer used as our SIEM, but the intern (Garry's mentee) was able to, through just memorized keyboard shortcuts and sounds, print out a few logs before the SIEM went down fully.

Figure out who was most likely to have done it given some of the login and Sysmon logs.

Note:
Record numbers have been modified after data was lost to be chronologically consistent & nothing else
Login GUIDs have been replaced by placeholders after a computer malfunction

Do not assume there is significant meaning behind Login GUID or record numbers

Printed Log Files Before System ShutDown:

10/23/2026 06:00:05 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4624
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Logon
OpCode=Info
RecordNumber=1512111014
Keywords=Audit Success
Message=An account was successfully logged on.

Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Logon Information:
    Logon Type: 3
    Restricted Admin Mode: -
    Virtual Account: No
    Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
    Security ID: S-1-5-21-1234567890-2345678901-3456789012-1001
    Account Name: Scientist
    Account Domain: DuckCorporation.local
    Logon ID: 0x9968D6B
    Linked Logon ID: 0x0
    Network Account Name: -
    Network Account Domain: -
    Logon GUID: {8E415A8E-7DE5-3ED8-5DDF-02AA51AE50E7}

Process Information:
    Process ID: 0x0
    Process Name: -

Network Information:
    Workstation Name: -
    Source Network Address: fe80::b574:557a:2d92:ce62
    Source Port: 51624

Detailed Authentication Information:
    Logon Process: Kerberos
    Authentication Package: Kerberos
    Transited Services: -
    Package Name (NTLM only):   -
    Key Length: 0

10/23/2026 06:01:00 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4634
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Logoff
OpCode=Info
RecordNumber=1512111020
Keywords=Audit Success
Message=An account was logged off.

Subject:
    Security ID: S-1-5-21-1234567890-2345678901-3456789012-1001
    Account Name: Scientist
    Account Domain: DUCKCORPORATION
    Logon ID: 0x9968D6B

Logon Type: 3

10/23/2026 06:04:07 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4624
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Logon
OpCode=Info
RecordNumber=1512111026
Keywords=Audit Success
Message=An account was successfully logged on.

Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Logon Information:
    Logon Type: 3
    Restricted Admin Mode: -
    Virtual Account: No
    Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
    Security ID: S-1-5-21-1234567890-2345678901-3456789012-1002
    Account Name: IT
    Account Domain: DuckCorporation.local
    Logon ID: 0x9968C34
    Linked Logon ID: 0x0
    Network Account Name: -
    Network Account Domain: -
    Logon GUID: {8E415A8E-7DE5-3ED8-5DDF-02AA51AE50E7}

Process Information:
    Process ID: 0x0
    Process Name: -

Network Information:
    Workstation Name: -
    Source Network Address: fe80::b574:557a:2d92:ce61
    Source Port: 51623

Detailed Authentication Information:
    Logon Process: Kerberos
    Authentication Package: Kerberos
    Transited Services: -
    Package Name (NTLM only):   -
    Key Length: 0

This event is generated when a logon session is created. It is generated on the
computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

The network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.

The impersonation level field indicates the extent to which a process in the logon session can impersonate.

The authentication information fields provide detailed information about this specific logon request.

- Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.
- Transited services indicate which intermediate services have participated in this logon request.
- Package name indicates which sub-protocol was used among the NTLM protocols.
- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4688
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Process Creation
OpCode=Info
RecordNumber=1512111036
Keywords=Audit Success
Message=A new process has been created.

Creator Subject:
    Security ID: [REDACTED/LOST BYTES]
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DUCKCORPORATION
    Logon ID: 0x809F9

Target Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Process Information:
    New Process ID: 0xe00
    New Process Name: C:\Windows\System32\winrs.exe
    Token Elevation Type: %%1936
    Mandatory Label: Mandatory Label\High Mandatory Level
    Creator Process ID: 0x17c0
    Creator Process Name: C:\Windows\System32\cmd.exe
    Process Command Line: winrs -r:Oxygen-Controller.DuckCorporation.local "Oxygen"

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

Type 1 is a full token with no privileges removed or groups disabled. A full token is only used if User Account Control is disabled or if the user is the built-in Administrator account or a service account.

Type 2 is an elevated token with no privileges removed or groups disabled. An elevated token is used when User Account Control is enabled and the user chooses to start the program using Run as administrator. An elevated token is also used when an application is configured to always require administrative privilege or to always require maximum privilege, and the user is a member of the Administrators group.

Type 3 is a limited token with administrative privileges removed and administrative groups disabled. The limited token is used when User Account Control is enabled, the application does not require administrative privilege, and the user does not choose to start the program using Run as administrator.

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10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=5156
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Filtering Platform Connection
OpCode=Info
RecordNumber=1512111043
Keywords=Audit Success
Message=The Windows Filtering Platform has permitted a connection.

Application Information:
    Process ID:     3584
    Application Name:   \device\harddiskvolume1\windows\system32\winrs.exe

Network Information:
    Direction:      Outbound
    Source Address:     10.0.1.14
    Source Port:        51627
    Destination Address:    10.0.1.15
    Destination Port:       5985
    Protocol:       6

Filter Information:
    Filter Run-Time ID: 68192
    Layer Name:     Connect
    Layer Run-Time ID:  48
10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=5158
EventType=0
Type=Information
ComputerName=Medical-Computer.DuckCorporation.local
TaskCategory=Filtering Platform Connection
OpCode=Info
RecordNumber=1512111042
Keywords=Audit Success
Message=The Windows Filtering Platform has permitted a bind to a local port.

Application Information:
    Process ID:     3584
    Application Name:   \device\harddiskvolume1\windows\system32\winrs.exe

Network Information:
    Source Address:     0.0.0.0
    Source Port:        51627
    Protocol:       6

Filter Information:
    Filter Run-Time ID: 0
    Layer Name:     Resource Assignment
    Layer Run-Time ID:  36
```

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4624
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Logon
OpCode=Info
RecordNumber=3702903
Keywords=Audit Success
Message=An account was successfully logged on.

Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Logon Information:
    Logon Type: 3
    Restricted Admin Mode: -
    Virtual Account: No
    Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
    Security ID: [REDACTED/LOST BYTES]
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DuckCorporation.local
    Logon ID: 0xA78EA19
    Linked Logon ID: 0x0
    Network Account Name: -
    Network Account Domain: -
    Logon GUID: {9825E01B-8303-8BF8-C4CD-5CD190000E66}

Process Information:
    Process ID: 0x0
    Process Name: -

Network Information:
    Workstation Name: -
    Source Network Address: -
    Source Port: -

Detailed Authentication Information:
    Logon Process: Kerberos
    Authentication Package: Kerberos
    Transited Services: -
    Package Name (NTLM only):   -
    Key Length: 0

This event is generated when a logon session is created. It is generated on the
computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

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The impersonation level field indicates the extent to which a process in the logon session can impersonate.

The authentication information fields provide detailed information about this specific logon request.

- Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.
- Transited services indicate which intermediate services have participated in this logon request.
- Package name indicates which sub-protocol was used among the NTLM protocols.
- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

10/23/2026 06:04:10 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4624
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Logon
OpCode=Info
RecordNumber=1512111048
Keywords=Audit Success
Message=An account was successfully logged on.

Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Logon Information:
    Logon Type: 3
    Restricted Admin Mode: -
    Virtual Account: No
    Elevated Token: Yes

Impersonation Level: Impersonation

New Logon:
    Security ID: S-1-5-21-1234567890-2345678901-3456789012-1003
    Account Name: Engineer
    Account Domain: DuckCorporation.local
    Logon ID: 0x9968D6C
    Linked Logon ID: 0x0
    Network Account Name: -
    Network Account Domain: -
    Logon GUID: {8E415A8E-7DE5-3ED8-5DDF-02AA51AE50E7}

Process Information:
    Process ID: 0x0
    Process Name: -

Network Information:
    Workstation Name: -
    Source Network Address: fe80::b574:557a:2d92:ce63
    Source Port: 51628

Detailed Authentication Information:
    Logon Process: Kerberos
    Authentication Package: Kerberos
    Transited Services: -
    Package Name (NTLM only):   -
    Key Length: 0

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4688
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Process Creation
OpCode=Info
RecordNumber=3702906
Keywords=Audit Success
Message=A new process has been created.

Creator Subject:
    Security ID: NT AUTHORITY\SYSTEM
    Account Name: Oxygen-Controller$
    Account Domain: DUCKCORPORATION
    Logon ID: 0x3E7

Target Subject:
    Security ID: NULL SID
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DUCKCORPORATION
    Logon ID: 0xA78EA19

Process Information:
    New Process ID: 0x68
    New Process Name: C:\Windows\System32\winrshost.exe
    Token Elevation Type: %%1936
    Mandatory Label: Mandatory Label\High Mandatory Level
    Creator Process ID: 0x2dc
    Creator Process Name: C:\Windows\System32\svchost.exe
    Process Command Line: C:\Windows\system32\WinrsHost.exe -Embedding

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

Type 1 is a full token with no privileges removed or groups disabled. A full token is only used if User Account Control is disabled or if the user is the built-in Administrator account or a service account.

Type 2 is an elevated token with no privileges removed or groups disabled. An elevated token is used when User Account Control is enabled and the user chooses to start the program using Run as administrator. An elevated token is also used when an application is configured to always require administrative privilege or to always require maximum privilege, and the user is a member of the Administrators group.

Type 3 is a limited token with administrative privileges removed and administrative groups disabled. The limited token is used when User Account Control is enabled, the application does not require administrative privilege, and the user does not choose to start the program using Run as administrator.

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4688
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Process Creation
OpCode=Info
RecordNumber=3702907
Keywords=Audit Success
Message=A new process has been created.

Creator Subject:
    Security ID: [REDACTED/LOST BYTES]
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DUCKCORPORATION
    Logon ID: 0xA78EA19

Target Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Process Information:
    New Process ID: 0xa34
    New Process Name: C:\Windows\System32\conhost.exe
    Token Elevation Type: %%1936
    Mandatory Label: Mandatory Label\High Mandatory Level
    Creator Process ID: 0x68
    Creator Process Name: C:\Windows\System32\winrshost.exe
    Process Command Line: \??\C:\Windows\system32\conhost.exe 0xffffffff -ForceV1

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

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Type 3 is a limited token with administrative privileges removed and administrative groups disabled. The limited token is used when User Account Control is enabled, the application does not require administrative privilege, and the user does not choose to start the program using Run as administrator.

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4688
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Process Creation
OpCode=Info
RecordNumber=3702908
Keywords=Audit Success
Message=A new process has been created.

Creator Subject:
    Security ID: [REDACTED/LOST BYTES]
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DUCKCORPORATION
    Logon ID: 0xA78EA19

Target Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Process Information:
    New Process ID: 0x580
    New Process Name: C:\Windows\System32\cmd.exe
    Token Elevation Type: %%1936
    Mandatory Label: Mandatory Label\High Mandatory Level
    Creator Process ID: 0x68
    Creator Process Name: C:\Windows\System32\winrshost.exe
    Process Command Line: C:\Windows\system32\cmd.exe /C "Oxygen"

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Type 3 is a limited token with administrative privileges removed and administrative groups disabled. The limited token is used when User Account Control is enabled, the application does not require administrative privilege, and the user does not choose to start the program using Run as administrator.

10/23/2026 06:04:12 PM
LogName=Security
SourceName=Microsoft Windows security auditing.
EventCode=4688
EventType=0
Type=Information
ComputerName=Oxygen-Controller.DuckCorporation.local
TaskCategory=Process Creation
OpCode=Info
RecordNumber=3702909
Keywords=Audit Success
Message=A new process has been created.

Creator Subject:
    Security ID: [REDACTED/LOST BYTES]
    Account Name: [REDACTED/LOST BYTES]
    Account Domain: DUCKCORPORATION
    Logon ID: 0xA78EA19

Target Subject:
    Security ID: NULL SID
    Account Name: -
    Account Domain: -
    Logon ID: 0x0

Process Information:
    New Process ID: 0xa84
    New Process Name: C:\Windows\System32\Oxygen.exe
    Token Elevation Type: %%1936
    Mandatory Label: Mandatory Label\High Mandatory Level
    Creator Process ID: 0x580
    Creator Process Name: C:\Windows\System32\cmd.exe
    Process Command Line: Oxygen

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

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Type 3 is a limited token with administrative privileges removed and administrative groups disabled. The limited token is used when User Account Control is enabled, the application does not require administrative privilege, and the user does not choose to start the program using Run as administrator.